

Enterprise ERP Project Documentation

1. Overview

ERP Suite using React + FastAPI + PostgreSQL with Local Storage, Amazon S3, and Google Cloud Storage.

2. Goals

Build enterprise-grade HR, Inventory, Finance, Vendor, Reporting, and Document Management modules.

3. Tech Stack

Frontend: React + TypeScript. Backend: FastAPI + Python + Uvicorn. Database: PostgreSQL. ORM: SQLAlchemy. Cache: Redis.

4. HLD

React UI communicates with FastAPI REST APIs. Backend handles auth, business logic, storage abstraction, and persistence.

5. Modules

Authentication, Employee Management, Inventory, Vendors, Finance, Reports, File Storage.

6. LLD

Layers: Router -> Service -> Repository -> Database. Schemas handled via Pydantic.

7. Database Schema

Core tables: users, roles, employees, products, vendors, invoices, files, audit_logs.

8. ER Relationships

Role 1:N Users. Department 1:N Employees. Employee 1:N Documents. Vendor 1:N Purchase Orders.

9. Folder Structure

app/main.py, routers/, services/, repositories/, schemas/, models/, storage/, utils/

10. Authentication

JWT access token + refresh token. Role-based access using middleware/dependencies.

11. File Upload Flow

React -> multipart/form-data -> FastAPI -> Storage Provider (LOCAL/S3/GCS) -> PostgreSQL metadata.

12. Storage Strategy

Abstract storage service with LocalStorageService, S3StorageService, GCSStorageService.

13. APIs

/auth/login, /employees, /inventory, /vendors, /files/upload, /reports.

14. S3 Integration

Use boto3 client, bucket erp-storage, store documents under employees/, invoices/, vendors/.

15. GCS Integration

Use google-cloud-storage SDK, bucket erp-gcs, blob upload for documents.

16. Security

bcrypt hashing, JWT, MIME validation, antivirus scan, HTTPS, audit logging.

17. Docker

Containers for React, FastAPI, PostgreSQL, Redis, Nginx.

18. CI/CD

Git -> Build -> Test -> Docker Build -> Deploy to cloud.

19. Monitoring

Prometheus + Grafana for API latency, error rate, storage usage.

20. Day-by-Day Implementation

Week1 Python/FastAPI basics; Week2 Auth; Week3 DB schema; Week4 CRUD; Week5 File Upload; Week6 S3/GCS; Week7 Reports; Week8 Deployment.

21. Interview Explanation

This ERP demonstrates scalable microservice-ready architecture with cloud storage abstraction and secure document management.